Adjusted Contracts for Price Risk Management

PhD Meeting

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Outline

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Motivation

- Agricultural commodities have experienced **substantial price volatility** in recent years
- Financial tools (forwards, futures, options, insurance) can mitigate price risk in the short run
- But adoption remains lower than expected from standard risk-averse utility maximization

Research Objective

Aim

Design and evaluate **adjusted contracts** for price risk management.

Specifically:

1. Introduce **price ceiling** for inputs and **price floor** for outputs
2. Evaluate contract performance via **stochastic simulation**
3. Compare outcomes under different:
   - Decision theories: EUT vs. CPT preferences
   - Framing: Broad framing (risk management) vs. narrow framing (investment)

Case study: Representative winter wheat farms in Germany

Decision Theories

Expected Utility Theory (EUT)

$$max\, CE = E[\pi] - \frac{r}{2} \cdot \mathrm{Var}[\pi]$$

Cumulative Prospect Theory (CPT)

$$max\, V = \sum_{i=1}^{n} \vartheta_i v\left(x_i\right)$$

$$v(x) = \begin{cases} (x_i - \mathrm{ref})^{\alpha} & \text{if } x_i \geq \mathrm{ref} \\ -\lambda(\mathrm{ref} - x_i)^{\alpha} & \text{if } x_i < \mathrm{ref} \end{cases}$$

$$w\left(p\right) = \exp\left\{ - \left[-\ln\left(p\right)\right]^{\gamma}\right\}$$

$$\vartheta_i^{+} = w\left(p_i + \ldots + p_n\right) - w\left(p_{i+1} + 1 + \ldots + p_n\right)$$

$$\vartheta_i^{-} = w\left(p_m + \ldots + p_i\right) - w\left(p_m + 1 + \ldots + p_{i-1}\right)$$

Decision Framing

Broad framing

- Contract as *risk management tool*
- Utility over total farm profits
- $\pi^{broad} = p_y \cdot y - p_x \cdot x + \pi^{contract}$
- Reference point: expected profit

Narrow framing

- Contract as *standalone investment*
- Utility over contract payoff only
- $\pi^{narrow} = \pi^{contract}$
- Reference point: zero

Contract Design

	Spot price	Forward	Input contract	Output contract
Effective price	p_x^s	p_x^f	$\min(p_N, k_N)$	$\max(p_w, k_w)$
Contract cost	–	–	Γ_N	Γ_w
Mechanism	–	Price fixing	Price ceiling	Price floor

Contract payoff:

$$I_N = \max(p_N - k_N, 0), \qquad I_w = \max(k_w - p_w, 0)$$

Contract premium:

$$\Gamma_i = E(I_i)e^{-rT}, \quad i = [N, w]$$

Methodology

1. **Production risk:** Just & Pope stochastic production function

$$y(N) = f(N) + \sigma_y(N)\,\varepsilon$$

$$f(N) = \beta_0 + \beta_1 N^{0.5} + \beta_2 N + e$$

$$\sigma_y(N) = |\hat{e}| = \beta_0 + \beta_1 N^{0.5}$$

2. **Price risk:** Quantile Vector Autoregression (QVAR) + Copula

$$p_N(q_N|\boldsymbol{P}_{t-1}) = \boldsymbol{\beta}_{0,N}(q_N) + \boldsymbol{\beta}_{1,N}(q_N)\,\boldsymbol{P}_{t-1}$$

$$p_w(q_w|\boldsymbol{P}_{t-1}) = \boldsymbol{\beta}_{0,w}(q_w) + \boldsymbol{\beta}_{1,w}(q_w)\,\boldsymbol{P}_{t-1}$$

$$C(q_w|q_N) = \boldsymbol{\beta}_{0(q_w)} + \boldsymbol{\beta}_{1(q_w)q_N}$$

3. **Contract pricing:** Burn analysis on simulated price paths
4. **Contract evaluation:** Monte Carlo simulation under EUT and CPT under broad and narrow framing

Data

Yield data

Table: Summary statistics of simulated wheat yield (t/ha) for different N levels from the EPIC-IIASA crop model

kgN/ha	Median	MAD*	Min	Max	Obs
10	1.6	0.18	1.16	2.31	35
50	3.63	0.69	2.02	4.65	140
100	5.76	0.87	3.05	7.15	140
180	7.13	1.32	4.30	9.43	324
250	11.2	0.86	4.68	13.3	210

*Median of Absolute Deviation (scaled by a correction factor of 1.4826)

Data

Price data

- Farm-level Calcium Ammonium Nitrate (CAN) and wheat prices (Chamber of Agriculture)
- 401 biweekly observations (Jan 2009 – Nov 2025)
- First differences (non-stationary) of log of prices

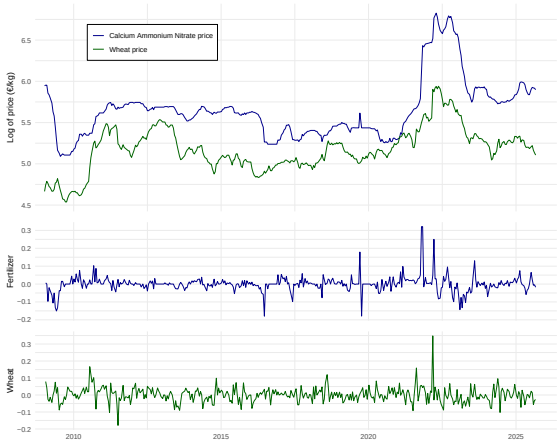


Figure: CAN and wheat prices (Jan 2009 – Nov 2025)

Results: Production Function

Table: Estimation of production and variation functions

	Yield function	Variation function
Intercept	-0.321	-0.109
	(0.205)	(0.122)
$N^{0.5}$	0.256^{***}	0.115^{***}
	(0.042)	(0.025)
N	0.028^{***}	-0.005^{***}
	(0.002)	(0.001)
Adj. R^2	0.936	0.170
N obs.	849	849

Notes: statistics in parentheses are standard errors; significance at * = 0.1, ** = 0.05, and *** = 0.01 levels.

Results: Copula estimation

Table: Quantile estimation of conditional copula

			q_w		
Quantile	0.1	0.3	0.5	0.7	0.9
Intercept	0.036	0.224***	0.520***	0.690***	0.868***
	(0.024)	(0.042)	(0.063)	(0.044)	(0.037)
q_N	0.148***	0.129**	-0.034	0.000	0.053
	(0.054)	(0.073)	(0.093)	(0.078)	(0.075)

Note: q_w and q_N denote the quantile distribution of wheat and CAN. Standard Errors in parenthesis are calculated by bootstrapping; stars indicate significance at * = 0.1, ** = 0.05, and *** = 0.01 levels.

Results: Contract Pricing

Contract premium (Γ) at different strike levels (k) by burn analysis:

$$\Gamma_i = \frac{1}{n} \sum_{t-1}^{n} (I_i)\, e^{-rT}, \quad i = [N, w]$$

Table: Input contract.

Quantile	Strike (€/t)	Premium (€/t)
0.5	352.47	12.50
0.6	359.55	9.38
0.7	367.97	6.47
0.8	377.32	4.22
0.9	390.60	2.31

Table: Output contract.

Quantile	Strike (€/t)	Premium (€/t)
0.1	128.38	0.873
0.2	136.79	2.081
0.3	144.07	3.819
0.4	149.90	5.805
0.5	157.20	8.966

Results: Broad Framing — EUT

Table: Certainty Equivalents (CE) for different level of absolute risk aversion (δ) under broad framing

r	forward contract	input contract 1	input contract 3	input contract 5	output contract 1	output contract 3	output contract 5
10^{-5}	1137.1	1139.4	1140.0	1139.5	1139.9	1140.6	**1141.6**
10^{-4}	1132.5	1134.9	1135.6	1135.1	1135.7	1136.9	**1138.4**
10^{-3}	1086.6	1089.8	1091.0	1090.9	1094.0	1099.8	**1106.6**
10^{-2}	882.5	889.7	892.7	894.6	908.9	935.3	**965.1**

- Wheat floor yields **highest CE** across all risk aversion levels

Results: Broad Framing — CPT

Table: CPT results under broad framing

	forward contract	input contract			output contract		
		1	3	5	1	3	5
$a = 0.334;\ \lambda = 1.574;\ \gamma = 0.571$	−1.93	−1.91	−1.89	−1.89	−1.87	−1.79	−1.69

- All prospect values are **negative** → CPT farmer would not voluntarily adopt
- Consistent across alternative parametrization

Narrow Framing — EUT

Table: Certainty Equivalents (CE) for different level of absolute risk aversion (δ) under narrow framing

r	forward contract	input contract 1	input contract 3	input contract 5	output contract 1	output contract 3	output contract 5
10^{-5}	**2.43**	−0.13	0.51	−0.03	0.21	0.94	2.17
10^{-4}	**2.29**	−0.19	0.47	−0.04	−0.09	−0.60	−1.65
10^{-3}	**1.81**	−0.39	0.35	−0.09	−1.07	−5.67	−14.17

- Forward contract **dominates** under narrow framing → farmer would not adopt adjusted contract

Narrow Framing — CPT

Table: CPT results under narrow framing

	forward contract	input contract			output contract		
		1	3	5	1	3	5
$a = 0.334;\ \lambda = 1.574;\ \gamma = 0.571$	-0.52	−0.65	−0.63	−0.78	−1.34	−1.44	−1.30
$\lambda = 0.5$	0.75	0.49	0.39	0.16	0.24	0.62	0.98

- Adjusted contracts are **still negative** → farmers not adopting
- Premiums trigger **loss aversion**
- Consistent across alternative parametrization; except loss seekers ($\lambda = 0.5$)

Summary of Results

	Broad framing	**Narrow framing**
EUT	Output contract attractive	Not attractive
CPT	Not attractive (all $V < 0$)	Not attractive (all $V < 0$)

Key insights:

- Under **broad framing** + **EUT**, addressing output price risk provides welfare gains
- Under **narrow framing**, contracts are **unattractive**

Discussion

Key takeaways:

- Framing matters: broad vs. narrow framing leads to **different conclusions** under EUT
- CPT consistently predicts **low adoption** – aligning with empirical observations
- Input price contracts are not attractive to farmers

Limitations:

- Single crop, single input simplification
- One period decision
- EUT and CPT results sensitive to parameter specification

Thank you!

Comparison of Contracts Design

	Fertilizer call	**Wheat put**	**Combined**
Type	Call option	Put option	Call + Put
Risk hedged	Input cost ↑	Revenue ↓	Both
Effective price	$\min(p_x^s, K_f)$	$\max(p_w^s, K_w)$	Both
Premium	ρ_f (/kg)	ρ_w (/t)	$\rho_f + \rho_w$
Strike quantiles	50th–90th	10th–50th	Diagonal pairing
Maturity	6 months	12 months	Both

Diagonal pairing: fertilizer strike i is paired with wheat strike i (rank-matched).

Full production and variation functions regression (1/2)

Table: Just & Pope production function estimates (robust MM regression).

	Yield function	Variation function
(Intercept)	-0.321	-0.109
	(0.205)	(0.122)
$\sqrt{\text{FTN}}$	0.256^{***}	0.115^{***}
	(0.042)	(0.025)
FTN	0.028^{***}	-0.005^{***}
	(0.002)	(0.001)
irr	0.060	-0.001
	(0.072)	(0.041)

Full production and variation functions regression (2/2)

	Yield function	Variation function
til_min_cons	−1.695***	0.040
	(0.088)	(0.053)
til_contill_r00	0.917***	−0.133***
	(0.082)	(0.049)
til_contill_r30	0.574***	0.029
	(0.106)	(0.060)
til_contill_r60	0.440***	0.225***
	(0.144)	(0.065)
til_contill_r90	0.031	0.406***
	(0.200)	(0.084)
Num. Obs.	849	849
R^2	0.937	0.170
R^2 Adj.	0.936	0.162
RMSE	1.17	0.87

Time series statistics and tests

Table: Summary statistics of log-prices and first differences.

Statistic	log(CAN)	log(Wheat)	ΔCAN	ΔWheat
Mean	5.65	5.17	0.00	0.00
SD	0.35	0.26	0.04	0.04
Min	5.09	4.54	−0.18	−0.18
Max	6.83	5.94	0.32	0.35
Skew	1.41	0.22	1.74	1.28
Kurtosis	5.26	3.56	20.20	13.43
ADF test	−2.76	−2.43	−7.19*	−8.69*
KPSS test	1.52*	1.63*	0.11	0.12
PP test	−10.51	−10.79	−264.99*	−309.15*

Note: statistics rounded to the second decimal of Calcium Ammonium Nitrate (CAN) and wheat prices; * indicates the rejection of the null hypothesis at the 5% level for the Augmented Dickey-Fuller (ADF) test, the Kwiatkowski-Phillips-Schmidt-Shin (KPSS), and the Phillips-Perron (PP) test.

Time series statistics and tests

Table: VAR model comparison: information criteria and serial correlation

Model	AIC	BIC	BG p-value
VAR(1)	−2808.33	−2776.44	0.044
VAR(1) + sq	−2824.01	−2776.18	0.191
VAR(1) + D2	−2806.12	−2766.25	0.039
VAR(1) + D_shock	−2785.58	−2745.80	0.087
VAR(1) + Q	−2821.82	−2766.01	0.099
VAR(1) + sq + D_shock	−2801.41	−2745.70	0.016
VAR(1) + sq + Q	−2833.68	−2761.93	0.148
VAR(1) + tt	−2785.06	−2737.31	0.109
VAR(2)	−2808.94	−2761.14	0.566
VAR(2) + sq	−2826.36	−2762.62	0.585
VAR(2) + D2	−2806.45	−2750.68	0.482
VAR(2) + sq + Q	−2836.34	−2748.70	0.347

Notes: AIC = Aikake Information Criteria; BIC = Bayesian Information Criteria. BG: Breusch–Godfrey LM test for serial correlation. sq = squared first lags ($\Delta f^2_{p,t-1}$, $\Delta w^2_{p,t-1}$); Q = quarterly dummies (Q1–Q3); D2 = level-shift dummy at ≈ 2021-01; D_shock = transitory shock dummy (Ukraine war); tt = time trends.

QVAR estimation (1/2)

Table: QVAR(1) estimates for CAN fertilizer price.

	VAR	QVAR				
		$\tau = 0.1$	$\tau = 0.3$	$\tau = 0.5$	$\tau = 0.7$	$\tau = 0.9$
Intercept	-0.000	-0.034^{***}	-0.009^{***}	0.000	0.009^{***}	0.034^{***}
	(0.002)	(0.004)	(0.002)	(0.001)	(0.001)	(0.006)
f_{p1}	0.384^{***}	0.445^{***}	0.318^{***}	0.303^{***}	0.288^{***}	0.261^{***}
	(0.045)	(0.082)	(0.018)	(0.019)	(0.045)	(0.088)
w_{p1}	0.203^{***}	-0.020	0.087^{***}	0.078^{***}	0.121^{***}	0.274^{**}
	(0.046)	(0.085)	(0.030)	(0.023)	(0.033)	(0.112)
Obs.	396	396	396	396	396	396
Adj. R^2	0.133					
Pseudo R^2		0.111	0.092	0.052	0.071	0.080
RMSE	0.04	0.05	0.04	0.04	0.04	0.05

Note: quantile regression standard errors are calculated by bootstrapping; stars indicate significance at * = 0.1, ** = 0.05, and *** = 0.01 levels; Pseudo R2 for quantile regression are calculated following Koenker & Machado (1999).

QVAR estimation (2/2)

Table: QVAR(1) estimates for wheat price.

	VAR	QVAR				
		$\tau = 0.1$	$\tau = 0.3$	$\tau = 0.5$	$\tau = 0.7$	$\tau = 0.9$
Intercept	0.001	-0.048^{***}	-0.016^{***}	0.000	0.016^{***}	0.045^{***}
	(0.002)	(0.004)	(0.002)	(0.002)	(0.002)	(0.005)
f_{p1}	0.100^{**}	0.242^{**}	0.146^{***}	0.134^{***}	0.084^{***}	0.037
	(0.048)	(0.098)	(0.033)	(0.027)	(0.028)	(0.108)
w_{p1}	0.226^{***}	0.069	0.227^{***}	0.228^{***}	0.229^{***}	0.246^{**}
	(0.049)	(0.100)	(0.050)	(0.046)	(0.054)	(0.108)
Obs.	396	396	396	396	396	396
Adj. R^2	0.061					
Pseudo R^2		0.033	0.051	0.044	0.036	0.026
RMSE	0.04	0.06	0.05	0.04	0.04	0.06

Note: quantile regression standard errors are calculated by bootstrapping; stars indicate significance at * = 0.1, ** = 0.05, and *** = 0.01 levels; Pseudo R2 for quantile regression are calculated following Koenker & Machado (1999).

Copula estimation

Table: Quantile estimation of conditional copula

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Intercept	0.036	0.224***	0.520***	0.690***	0.868***
	(0.024)	(0.042)	(0.063)	(0.044)	(0.037)
q_N	0.148***	0.129**	-0.034	0.000	0.053
	(0.054)	(0.073)	(0.093)	(0.078)	(0.075)

Note: q_w and q_N denote the quantile distribution of wheat and CAN. Standard Errors in parenthesis are calculated by bootstrapping; stars indicate significance at * = 0.1, ** = 0.05, and *** = 0.01 levels.

Profits calculation

Table: Profit formula for different contracts

Contract	**Profit formula**
No contract	$p_w^s \cdot y - p_x^s \cdot \text{CAN}$
Forward (fertilizer)	$p_w^s \cdot y - p_x^f \cdot \text{CAN}$
Fert. ceiling only	$p_w^s \cdot y - [\min(p_x^s, K_f) + \rho_f] \cdot \text{CAN}$
Wheat floor only	$[\max(p_w^s, K_w) - \rho_w] \cdot y - p_x^s \cdot \text{CAN}$

- Evaluated under **1,000 simulated price paths** (copula-linked)
- Broad framing: ref $= E[\pi^{fc}]$ | Narrow framing: ref $= 0$

Profit Summary Statistics (€/ha)

Table: Summary statistics of profits for different contracts

	Mean	Median	SD	Min	Max
No contract (nc)	1140.01	1096.79	311.50	272.42	2296.22
Forward (fc)	1137.58	1099.02	319.41	255.29	**2385.43**
ac_1 (fert. ceiling)	1139.88	1097.88	316.35	264.28	2380.36
ac_5	1139.99	1097.83	313.30	270.94	2362.58
wc_1 (wheat floor)	1140.31	1092.97	304.24	**382.24**	2287.53
wc_5	**1141.91**	**1117.20**	**265.93**	420.25	2207.99